Uppsala University



Machine Learning

Assignment 5

General information

- The recommended tool in this course is R (with the IDE R-Studio). You can download R **here** and R-Studio **here**. You can use Python and Jupyter Notebooks, although the assignments may use data available only through the R package, a problem you would need to solve yourself.
- Report all results in a single, `*.pdf`-file. *Other formats, such as Word, Rmd, Jupyter Notebook, or similar, will automatically be failed.* Although, you are allowed first to knit your document to Word or HTML and then print the assignment as a PDF from Word or HTML if you find it difficult to get TeX to work.
- The report should be written in English.
- If a question is unclear in the assignment. Write down how you interpret the question and formulate your answer.
- You should submit the report to **Studium**. Deadlines for all assignments are **Sunday 23.59**. See **Studium** for dates. Assignments will be graded within 10 working days from the assignment deadline.
- To pass the assignments, *you should answer all questions not marked with **, and get at least 75% correct.
- To get VG on the assignment, *also the questions marked with a ** should be answered. Sometimes you can choose between different VG assignments, then this is explicitly stated. To get VG you need 75% both on questions to pass and the VG part of the assignment. VG will only be awarded on the first deadline of the assignment.
- A report that does not contain the general information (see the **template**), will be automatically rejected.
- Every assignment report must include one oral exam question of your own together with a short model answer, following the oral exam question contribution instructions in this assignment. You are not allowed to use large language models, such as ChatGPT, at all for this contribution.
- State clearly whether the course staff may use or adapt your question and answer in a future oral exam: **Yes** or **No**.
- When working with R, we recommend writing the reports using R markdown and the provided **R markdown template**. The template includes the formatting instructions and how to include code and figures.
- Instead of R markdown, you can use other software to make the pdf report, but you should use the same instructions for formatting. These instructions are also available in **the PDF produced from the R markdown template**.
- The course has its own R package `uuml` with data and functionality to simplify coding. To install the package just run the following:

```
1. install.packages("remotes")
```

2. `remotes::install_github("MansMeg/IntroML", subdir = "rpackage")`

- We collect common questions regarding installation and technical problems in a course Frequently Asked Questions (FAQ). This can be found **here**.
- You are not allowed to show your assignments (text or code) to anyone. Only discuss the assignments with your fellow students. The student that show their assignment to anyone else could also be considered to cheat. Similarly, on zoom labs, only screen share when you are in a separate zoom room with teaching assistants.
- The computer labs are for asking all types of questions. Do not hesitate to ask! The purpose of the computer labs are to improve your learning. We will hence focus on more computer labs and less on assignment feedback. *Warning!* There might be bugs in the assignments! Hence, it is important to ask questions early on so you dont waste time of unintentional bugs.
- If you have any suggestions or improvements to the course material, please post in the course chat feedback channel, create an issue **here**, or submit a pull request to the public repository.

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Oral exam question contribution

Write one possible oral exam question of your own for this assignment together with a short model answer. Include both in your assignment report.

The question should:

- concern material from this assignment or the associated reading or lectures,
- be suitable for a 2–4 minute oral answer, possibly using a whiteboard,
- test conceptual understanding rather than only calculation or code,
- not already be included in the oral exam question list and not be very close to an existing question (e.g. just different numbers).

Note! You are not allowed to use large language models, such as ChatGPT, at all for this oral exam question contribution.

State clearly whether the course staff may use or adapt your question and answer in a future oral exam: **Yes** or **No**.

1 Transformers and Attention

In this assignment, we will implement a multi-head attention transformer layer. These layers are currently used within all transformer-based models, such as GPT and different BERT models. A good read for this assignment is the Illustrated transformer by Jay Alammar, which you can find [here](http://jalammar.github.io/illustrated-transformer/).

Start by loading the parameters that we are going to use for the implementation as follows.

```
library(uuml)
data("transformer_example")
```

1. (1p) Start out by computing the query, key and value matrices for one attention head by implementing it in a function you call `qkv()`. The function should work like this.

```
# Pick out the matrix for the first attention head
Wq <- transformer_example$Wq[,,1]
Wk <- transformer_example$Wk[,,1]
Wv <- transformer_example$Wv[,,1]

# Pick out the first three words and their embeddings
X <- transformer_example$embeddings[1:3,]

qkv(X, Wq, Wk, Wv)

## $Q
##             [,1]        [,2]       [,3]
## the    0.4722259  0.04995783 -0.5350845
## quick -0.3662435  0.12144160  0.3454785
## brown -0.1029677 -0.12728414  0.1817097
##
## $K
##                [,1]          [,2]       [,3]
## the    0.094360579 -0.203807092 -0.1851229
## quick -0.033313240  0.279012100  0.2530560
## brown -0.004457052  0.001013468  0.0133802
##
## $V
##                [,1]        [,2]        [,3]
## the    0.317318525 -0.35023010  0.13284078
## quick  0.009929565  0.04208206 -0.15412097
## brown -0.316413241  0.27717408  0.02725089
```

2. (2p) Now, based on your query, key and value, compute the attention of that given attention head for the three chosen tokens.

```
attention(res$Q, res$K, res$V)
## $Z
##                [,1]          [,2]         [,3]
```

```
## the    0.012395453 -0.0212420459  0.009404870
## quick -0.003759269 -0.0008360029 -0.005108890
## brown  0.002412222 -0.0088974612  0.001147999
##
## $attention
##              the     quick     brown
## the   0.3601932 0.3080896 0.3317172
## quick 0.3088780 0.3582373 0.3328847
## brown 0.3300375 0.3360583 0.3339042
```

3. (1p) Interpret the attention values. What does the second row mean?

4. (2p) Now we have everything in place for implementing it all as a multi-head attention layer. The layer will take in embeddings and then return a 3-dimensional embedding per word. Run your code on all words in the included example.

```
multi_head_self_attention(X,
                          transformer_example$Wq,
                          transformer_example$Wk,
                          transformer_example$Wv,
                          transformer_example$W0)
##               [,1]          [,2]         [,3]
## the   -0.014189613 -0.0040299008 -0.006756286
## quick -0.009963516 -0.0010724342 -0.001996524
## brown -0.006394562 -0.0006626115 -0.002219108
```

2 Implementing a simple RNN

We are going to implement a one-layer recurrent neural network based on the `rnn_example` data. We are going to implement a simple layer described in Section 10.2 in Goodfellow et al. (2016). See this section for details.

We will set the dimensionality of the hidden state to 4 and the output dimension to 3. In this task, we will set h_0, the initial starting state, to a zero-vector. The input will be the embeddings in the `rnn_example` data.

```
library(uuml)
data("rnn_example")
```

1. (2p) We start out by implementing a simple RNN linear unit that takes the parameters `W`, `U`, `b`, the previous hidden state `h_t_minus_one` and an input embedding `x_t` and return the activation input a_t. It should work as follows.

```
X <- rnn_example$embeddings[1,,drop=FALSE]
h_t_minus_one <- matrix(0, nrow = hidden_dim, ncol = 1)
a_t <- rnn_unit(h_t_minus_one, X,
                W = rnn_example$W,
                U = rnn_example$U,
                b = rnn_example$b)
a_t
##
##              the
## [1,]  0.5819145
## [2,] -2.2686535
## [3,] -0.6410312
## [4,]  1.4891931
##
```

2. (1p) Now implement the `tanh()` activation function.

```
h_t <- activation(a_t)
h_t
##
##              the
## [1,]  0.5240555
## [2,] -0.9788223
## [3,] -0.5656013
## [4,]  0.9031762
##
```

3. (1p) As the next step, implement the output function and the softmax function. These functions should work in the following way.

```
yhat_t <- softmax(output_rnn(h_t, rnn_example$V, rnn_example$c))
yhat_t
##
```

```
##             the
## [1,] 0.3063613
## [2,] 0.2930885
## [3,] 0.4005502
##
```

4. (2p) Now we are ready to implement the full recurrent layer. It should take an input matrix X and the neural network parameters and return the hidden states and the softmax output as follows.

```
X <- rnn_example$embeddings[1:3,,drop=FALSE]
rnn_layer(X,
          W = rnn_example$W,
          V = rnn_example$V,
          U = rnn_example$U,
          rnn_example$b,
          rnn_example$c)
##
## $ht
##               [,1]        [,2]       [,3]       [,4]
## the    0.52405551 -0.97882227 -0.5656013  0.9031762
## quick -0.05951368  0.03988226  0.8241800 -0.6562744
## brown -0.08984008  0.92822217 -0.1563247 -0.6657626
##
## $yhat
##             [,1]      [,2]      [,3]
## the   0.3063613 0.2930885 0.4005502
## quick 0.2838013 0.3490452 0.3671536
## brown 0.2878002 0.3666877 0.3455121
```

5. (1p) Now, what is the value of the hidden state `h_t` for the token `dog`?

3 *Recurrent Neural Networks with Keras

To get VG on this assignment, you can choose to do this task OR the task using BERT with PyTorch below. You are also free to do both, if you want. As long as you pass one of the VG assignments you will get a VG point. Although, please only write down the time it took to do the VG assignment for one of the tasks if you choose to do more than one.

This task is only necessary to get one point toward a *pass with distinction* (VG) grade. Hence, if you do not want to get a *pass with distinction* (VG) point, you can ignore this part of the assignment.

As a first step, we will try using Recurrent Neural Networks on a text classification task. Here Chapter 6 - 6.2 in Chollet and Allaire (2018) will be the primary reference.

Start by downloading the IMBD dataset and preprocess it as follows.

```
library(tensorflow)
library(keras)
imdb <- dataset_imdb(num_words = 10000)
c(c(input_train, y_train), c(input_test, y_test)) %<-% imdb
input_train <- pad_sequences(input_train, maxlen = maxlen)
input_test <- pad_sequences(input_test, maxlen = maxlen)
```

Note! Running Keras can be computationally heavy, and I suggest not to run the code in `markdown`. Instead, run the code in R and copy the results as output (see the assignment template for an example).

1. (2p) Create a simple ordinary neural network for text classification in Keras using an embedding layer of dimension 16 and a hidden layer with 32 hidden states (and an output layer with one unit). Use a validation split of 20% observations in the validation set and a batch size of 128. Train your network using rmsprop and use binary cross-entropy as the loss function. Print the model and return your validation accuracy after ten epochs. Most of these settings we will reuse later in the assignments.

2. (1p) Explain what the embedding layer is doing.

3. (1p) Now setup a simple Recurrent Neural Network with 32 hidden units based on the same embedding layer. Print the model and present your result.

4. (1p) The model you have implemented now. Which recurrent network does best describes the model you have now implemented out of Fig. 10.3 and 10.5 in Goodfellow et al. (2016)? Motivate why.

5. (1p) Explain the role of the parameters U, W in the recurrent net (see Goodfellow et al., 2017 Section 10.2). How large are these parameters in your model specified above, and where are they included in the Keras output?

6. (1p) Now, explain the role of the parameters V (see Goodfellow et al., 2017 Section 10.2). How many parameters are included in V, and where are they included in the Keras output?

7. (1p) We are now going to implement an LSTM network. Implement a standard LSTM network with 32 hidden units using Keras. Print your model and your validation accuracy.

8. (1p) Extend your network in ways you think fit. Report at least two other extensions/modifications of your network and why you chose them. Report the Keras model output and validation accuracy.

9. (1p) Reason why we cannot get as good performance as with the MNIST dataset. Is the problem bias, variance, or the Bayes error? Why?

4 *BERT for Swedish with PyTorch

To get VG on this assignment, you can choose to do this task OR the task using Keras and PyTorch above. You are also free to do both if you want. You will get a VG point if you pass one of the VG assignments. Although, please only write down the time it took to do the VG assignment for one of the tasks if you choose to do more than one.

This task is only necessary to get one point toward a *pass with distinction* (VG) grade. Hence, if you do not want to get a *pass with distinction* (VG) point, you can ignore this part of the assignment.

Unfortunately, there is yet to be a simple way to run BERT directly from R. Although, using Google Colab, we can run one of the Swedish BERT models trained by the Swedish national library. The tutorial is in English, and without any Swedish understanding, it should be straightforward to follow.

Follow the tutorial on Google Colab here. Then do the following things.

1. (2p) Try to improve on the relatively low classification accuracy and get above 80% overall accuracy. You are free to elaborate how much you want yourself, but some suggestions are:

   (a) Fine-tune a larger part of the BERT model (not just the classification head)

   (b) Run the model for more epochs

   (c) Try out other learning rates (scheduling)

   Note! Training these large neural networks might take some time. It can take more than an hour at Google Colab to fine-tune the whole BERT model.

2. (1p) Write down the steps you take, the overall accuracy you get along the way, and the final accuracy of your fine-tuned BERT model, both overall and per party.

3. (2p) Reason based on your final confusion matrix on the difficulties for the model. Can the errors be reduced further or not? Why?

References

Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016. http://www.deeplearningbook.org.

François Chollet and Joseph J Allaire. *Deep Learning with R*. Manning, 2018.